Jianye Xu

Ph.D. Student

Chair of Embedded Software, Department of Computer Science, RWTH Aachen University

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Research interests: Safe learning and distributed control, multi-agent decision making, and motion planning for connected and automated vehicles.

Education

RWTH Aachen University, Germany **2023-04 – 2027-03 (expected)**

- Ph.D., Chair of Embedded Software, Department of Computer Science
- *Thesis:* Learning Safe Coordination for Connected Vehicles (working title)
- *Advisor:* Prof. Bassam Alrifaee

RWTH Aachen University, Germany **2020-10 – 2022-09**

- M.Sc., Automation Engineering, Institute of Automatic Control
- Overall average: 1.2/1.0 · *with distinction*
- *Thesis:* Parallel Priority-Based Trajectory Planning with Safety Guarantees for Networked Vehicles
- *Advisor:* Prof. Stefan Kowalewski

Beijing Institute of Technology, China **2016-10 – 2020-09**

- B.Sc., Vehicle Engineering, School of Mechanical Engineering
- Overall average: 91.48/100 · *Outstanding Graduate (top 5%)*
- Final year as exchange student at RWTH Aachen University (2019-10 – 2020-08)
- *Thesis (at RWTH Aachen University):* Implementation and Comparison of Different Strategies for MTPA Control of Permanent-Magnet Synchronous Machines Considering Magnetic Saturation
- *Advisors:* Prof. Chen Lin (Beijing Institute of Technology), Prof. Jakob Andert (RWTH Aachen University)

Selected Publications

Journal Papers

[J1] J. Beerwerth, **J. Xu**, S. Schäfer, F. Belderink, and B. Alrifaee. "Zero-Shot MARL Benchmark in the Cyber-Physical Mobility Lab". In: *at - Automatisierungstechnik* 74.5 (2026), pp. 376–385. DOI: 10.1515/auto-2025-0057.

[J2] **J. Xu**, J. Betz, A. Mokhtarian, A. Mittal, M. Cai, R. Mangharam, O. M. Shehata, C. M. Elias, J.-N. Zaech, P. Scheffe, F. Jahncke, S. S. Ulhas, K. M. Arfvidsson, and B. Alrifaee. "Small-Scale Testbeds for Connected and Automated Vehicles and Robot Swarms: Challenges and a Roadmap". In: *IEEE Intelligent Transportation Systems Magazine* (2026), pp. 2–16. DOI: 10.1109/MITS.2026.3666681.

[J3] A. Mokhtarian, **J. Xu**, P. Scheffe, M. Kloock, S. Schäfer, H. Bang, V.-A. Le, S. Ulhas, J. Betz, S. Wilson, S. Berman, L. Paull, A. Prorok, and B. Alrifaee. "A Survey on Small-Scale Testbeds for Connected and Automated Vehicles and Robot Swarms: A Guide for Creating a New Testbed". In: *IEEE Robotics & Automation Magazine* (2024), pp. 2–19. DOI: 10.1109/MRA.2024.3505772.

Conference Papers

[C1] **J. Xu** and B. Alrifaee. "Beyond Safety Filtering: Control Barrier Function-Informed Reinforcement Learning for Connected and Automated Vehicles". In: *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC), in Press*. 2026.

[C2] **J. Xu** and B. Alrifaee. "A Learning-Based Control Barrier Function for Car-Like Robots: Toward Less Conservative Collision Avoidance". In: *2025 European Control Conference (ECC)*. 2025, pp. 988–995. DOI: 10.23919/ECC65951.2025.11187043.

[C3] **J. Xu**, C. Che, and B. Alrifaee. "A Real-Time Control Barrier Function-Based Safety Filter for Motion Planning with Arbitrary Road Boundary Constraints". In: *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*. 2025, pp. 2818–2825. DOI: 10.1109/ITSC60802.2025.11423203.

[C4] P. Scheffe, **J. Xu**, and B. Alrifaee. "Limiting Computation Levels in Prioritized Trajectory Planning with Safety Guarantees". In: *2024 European Control Conference (ECC)*. 2024, pp. 297–304. DOI: 10.23919/ECC64448.2024.10591179.

[C5] **J. Xu**, P. Hu, and B. Alrifaee. "SigmaRL: A Sample-Efficient and Generalizable Multi-Agent Reinforcement Learning Framework for Motion Planning". In: *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*. 2024, pp. 768–775. DOI: 10.1109/ITSC58415.2024.10919918.

Manuscripts Under Review

[U1] **J. Xu** and B. Alrifaee. "TTCBF: A Truncated Taylor Control Barrier Function for High-Order Safety Constraints". 2026. arXiv: 2601.15196. Under revision.

[U2] **J. Xu**, Y. Sayed, and B. Alrifaee. "Safeguarding Multi-Agent Reinforcement Learning for Connected and Automated Vehicles with Recursively Feasible Control Barrier Functions". 2026. arXiv: 2611.04957. Under review at IEEE Transactions on Intelligent Transportation Systems.

Awards

Best Poster Award German Autonomy Summit (GAS) **2025-12**

Best Paper Nomination IEEE International Conference on Intelligent Transportation Systems (ITSC) **2025-11**

Dean's List (top 5% in the academic year 2021/2022) RWTH Aachen University **2023-01**

M.Sc. with distinction (GPA 1.2/1.0) RWTH Aachen University **2022-09**

German National Scholarship (Deutschlandstipendium) RWTH Aachen University **2021-11**

Patent: Quick Locking and Unlocking Device for Vehicle-Mounted Power Battery Box (first inventor, publication number: CN 111086379 B) China National Intellectual Property Administration **2021-06**

German National Scholarship (Deutschlandstipendium) RWTH Aachen University **2020-11**

Outstanding Graduate (GPA 91.48/100, top 5%) Beijing Institute of Technology **2020-06**

China National Scholarship Ministry of Education of the People's Republic of China **2018-11**

Second Prize in National Student Mechanical Product Digital Design Competition China Mechanical Discipline Steering Committee **2018-06**

Outstanding Undergraduate Student Beijing Institute of Technology **2017, 2018, 2019**

First Prize in Beijing Student Engineering Design Expression Competition Beijing Municipal Commission of Education **2017-12**

First Prize in National Student Drafting and Modelling Innovation Competition China Cartographic Association **2017-07**

Second Prize in Engineering Drafting Skills Competition Beijing Institute of Technology **2017-06**

Professional Activities

Program Committee

Associate Editor **2026**
IEEE International Conference on Intelligent Transportation Systems (ITSC)

Invited Session Organizer *IEEE ITSC, Naples, Italy* **2026-09**
- Invited Session: "Bridging Optimal Control and Learning for Connected and Automated Vehicles"
- https://bassamlab.github.io/itsc2026-invited-session/

Workshop Co-Organizer *IEEE ITSC, Gold Coast, Australia* **2025-11**
- Workshop: "Automated Mobility in Mixed Traffic—Emerging Mixed Autonomy in the Era of AI"
- https://www.itsc2025.mixedtraffic.org/

Workshop Co-Organizer *IEEE IV, Jeju, South Korea* **2024-06**
- Workshop: "1st Workshop on Small-Scale Testbeds for Connected and Automated Vehicles and Robot Swarms"
- https://cpm.lrt.unibw.de/workshop/

Competition Co-Organizer *RWTH Aachen University, Germany* **2024-01 – 2024-06**
- Competition: CPM Olympics (a motion planning competition for connected and automated vehicles)
- https://cpm.lrt.unibw.de/olympics/

Invited Talks

Workshop Invited Speaker *IEEE ITSC, Naples, Italy* **2026-09**
- Workshop: "The 16th Workshop & Industry Panel on Cooperative and Automated Driving and Future Mobility"
- https://ieee-itsc.org/2026/workshops/

Workshop Invited Speaker *IEEE ITSC, Gold Coast, Australia* **2025-11**
- Workshop: "Automated Mobility in Mixed Traffic—Emerging Mixed Autonomy in the Era of AI"
- https://www.itsc2025.mixedtraffic.org/

Journal Reviewing

Automatica, IEEE Control Systems Letters (L-CSS), IEEE Transactions on Intelligent Transportation Systems (T-ITS), IEEE Intelligent Transportation Systems Magazine (ITSM), IEEE Robotics and Automation Letters (RA-L)

Conference Reviewing

IEEE International Conference on Intelligent Transportation Systems (ITSC), IEEE Intelligent Vehicles Symposium (IV), European Control Conference (ECC), IEEE International Conference on Robotics and Automation (ICRA)

Membership

IEEE Graduate Student Member, IEEE Intelligent Transportation Systems Society, IEEE Robotics and Automation Society, IEEE Control Systems Society, IEEE Young Professionals

Teaching and Mentoring

Assistant Lecturer *RWTH Aachen University, Germany* **Winter terms 2024 & 2025**

- Master's course: "Control and Perception in Networked and Autonomous Vehicles"
- https://cpm.lrt.unibw.de/courses/master/

Guest Lecturer *RWTH Aachen University, Germany* **2025-01**

- Guest Lecturer for Master's course: "Control Barrier Functions and Their Applications to Collision Avoidance in Networked and Autonomous Vehicles"

Undergraduate Teaching Assistant *Beijing Institute of Technology, China* **2017-09 – 2017-10**

- Delivered CAD software training to first-year students.

Supervised Theses *RWTH Aachen University* **2023 – 2026**

- Ali Elbadawy (2026). *Integrating Barrier Functions into Multi-Agent Social Value Orientation for Unsignalized Intersections* (Master's thesis).
- Massimo Marcon (2026). *Latent World Models for Motion Planning of Connected and Automated Vehicles* (Master's thesis).
- Jiaxin Liu (2026). *Mixture of Expert Policies for Decision-Making and Motion Planning in Connected and Automated Vehicles* (Master's thesis).
- George Halim (2026). *Barrier-Certified Exploration for Real-World Reinforcement Learning in the Cyber-Physical Mobility Lab* (Bachelor's thesis).
- Yousef Sayed (2026). *A Study of Responsibility Assignment in Control Barrier Function for Connected and Automated Vehicles* (Bachelor's thesis).
- Mohamed ElShaarawy (2026). *Large Language Model-Based Traffic Scenario Analysis and Decision-Making in the Cyber-Physical Mobility Lab* (Bachelor's thesis).
- Maximilian Petri (2025). *Reinforcement Learning for Formula Student Autonomous Racing* (Master's thesis).
- Chang Che (2025). *Constraint-Enforced Multi-Agent Reinforcement Learning for Safe Motion Planning of Connected and Automated Vehicles* (Master's thesis).
- Pan Hu (2025). *Situation-Aware Trajectory Planning of Connected and Automated Vehicles with Multi-Policy Reinforcement Learning* (Master's thesis).
- Daniel Boules (2025). *A Prioritization-Based Approach for Enhanced Stability in Non-Stationary Multi-Agent Reinforcement Learning* (Bachelor's thesis).
- Fynn Belderink (2025). *Model-Based Receding-Horizon Planning in Multi-Agent Reinforcement Learning* (Bachelor's thesis).
- Yunjia Chen (2025). *Sensor Fusion-Enhanced Map Matching of Bicycle Routes Using Crowdsourced GPS and IMU Data from Smartphones* (Bachelor's thesis).
- Xinyue Cui (2024). *Determining the Longitudinal and Lateral Acceleration of a Bicycle Using a Smartphone Mounted on Its Handlebar* (Master's thesis).
- Lebo Liang (2024). *Safe Multi-Agent Control: Integrating Control Barrier Functions with Reinforcement Learning* (Bachelor's thesis, Aachen University of Applied Sciences).
- Mohamed Shamekh (2024). *Model-Predictive-Control-Based Safe Multi-Agent Reinforcement Learning for Motion Planning* (Bachelor's thesis).
- Omar Sobhy (2024). *Dealing with Non-Stationarity in Multi-Agent Reinforcement Learning Through Learning-Based Dynamic Prioritization* (Bachelor's thesis).
- Xia Yan (2024). *Learning-Based Motion Planner for Connected and Automated Vehicles* (Master's thesis).
- Xilei Chen (2024). *Imitation Learning from Optimization-Based Motion Planners for Connected and Automated Vehicles* (Bachelor's thesis).
- Raphael Malig (2024). *Investigating the Effect of Discretizing Continuous Action Space in Multi-Agent Reinforcement Learning for Motion Planning* (Bachelor's thesis).

- Andreas Pletschko (2024). *Leveraging Temporal Dependencies for Sample-Efficient and Generalizable Multi-Agent Reinforcement Learning in Motion Planning* (Bachelor's thesis).
- Ruizhang Zhou (2023). *AI-Based Generation of Testing Scenarios for Motion Planners on Connected and Automated Vehicles* (Master's thesis).

Supervised Seminar Works *RWTH Aachen University* **2023 – 2026**

- Supervised seminar works in both summer and winter terms.